

19/11/25

Machine Learning Assignment - 5

Q1

- 1) Decision trees offer key advantages such as interpretability - their structure is intuitive and easy to visualize.

They handle both categorical and numerical data, function well with missing values, and are robust to outliers.

Decision Trees do not rely on assumptions about the underlying data distribution, so they can be applied to many types of problems.

Disadvantages:

- Overfitting: Trees can become overly complex, fitting minor noise ~~at~~ of the training data, resulting in poor generalization to the unseen data.
- Instability: Small changes in the training data can result in a completely different tree structure, making results unpredictable and non-reliable.

- Bias toward Dominant Classes: In cases of imbalanced data, trees may favour the most frequent class and ignore minority classes, reducing predictive power.
- Computational Expense and Scalability: For large datasets with many features, training decision trees can be computationally demanding, and the search space for splits increases exponentially.
- Greedy Nature: Decision trees build using greedy algorithms optimizing for the best local split, which may not lead to a globally optimal tree.

2) Some important hyperparameters:

- max_depth: Controls the maximum depth of the tree, limiting how many sequential splits can occur. Smaller values prevent overfitting but may underfit if too restrictive.
- min_samples_split: The minimum split number of samples required to split an internal node. Higher values prevent the tree from learning overly specific patterns and reduce overfitting.

- min-samples-leaf: The minimum no. of samples req. to be at a leaf node. This smooths the model by ensuring predictions are made based on sufficient data points.
- max-features: The number of features to consider when looking for the best split. Reducing this adds randomness and can prevent overfitting, particularly useful in ensemble methods.
- cp (complexity parameter): The cost-complexity pruning parameter that penalizes tree complexity. Higher values lead to more aggressive pruning and simpler trees.
- criterion: The function to measure split quality, such as Gini impurity or entropy for classification, and MSE or MAE for regression.

3) Overfitting a decision tree:

A decision tree overfits when its complexity allows it to model not just underlying trends, but also, noise specific to the training data, leading to poor generalization on new data. Three ways to overfit are:

- a) Allow unlimited tree depth: setting 'max-depth' to none or a very high value enables the tree to keep splitting until all leaves are pure causing it to memorize the training data instead of learning generalizable patterns.
- b) Very small min-samples-split/min-samples-leaf: Allowing splits with very few samples creates leaves that focus on ~~minute~~ outliers in the data, capturing noise.
- c) Fail to apply pruning or regularization: Not employing tree pruning or constraints leaves the tree unchecked, exaggerating complexity and vulnerability to noise in the training set.

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Team - ID - 47

Students: Juan, Alvarez, Prathul Ravuri,
Prathav GuptaML - Exercise - 05Qn-2) Classification and Regression Trees (CART): Hands-On1) $N=5$, $K=3$, $V = \{\text{Tennis, Football, Basketball}\}$, $P(V) = \{\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\}$

(i) Best initial split

$$N \cdot H(V) = 5 \cdot [-0.4 \cdot \log_2 0.4 - 0.4 \log_2 0.4 - 0.2 \log_2 0.2]$$

$$= 5 \cdot [1.52] = 7.6$$

Possible Splits: [Let Ball Color be X_1 and Ball Width be X_2]

① Ball Color: $\rightarrow$ (i) $L = \{X_1^{\text{Yellow}}\}$, $R = \{X_1^{\text{White, Brown}}\}$ $\rightarrow P(V^L) = \{1, 0, 0\}$, $N^L = 2$
 $P(V^R) = \{0, \frac{2}{3}, \frac{1}{3}\}$, $N^R = 3$

(ii) $L = \{X_1^{\text{White}}\}$, $R = \{X_1^{\text{Yellow, Brown}}\}$ $\rightarrow P(V^L) = \{0, 1, 0\}$, $N^L = 1$
 $P(V^R) = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\}$, $N^R = 4$

(iii) $L = \{X_1^{\text{Brown}}\}$, $R = \{X_1^{\text{Yellow, White}}\}$ $\rightarrow P(V^L) = \{0, \frac{1}{2}, \frac{1}{2}\}$, $N^L = 2$
 $P(V^R) = \{\frac{2}{3}, \frac{1}{3}, 0\}$, $N^R = 3$

Impurity Gains: $[N \cdot H(V) - N^L \cdot H(V^L) - N^R \cdot H(V^R)]$

I. i) $I = 7.6 - [2 \times 0 + 3 \times (-0.67 \cdot \log 0.67 - 0.33 \cdot \log 0.33)]$
 $= 7.6 - [0 + 2.74] = 4.86$

I. ii) $I = 7.6 - [1 \times 0 + 4 \times (-0.5 \log 0.5 - 2 \times 0.25 \log 0.25)]$
 $= 7.6 - [0 + 4 \times 0.5] = 1.6$

I. iii) $I = 7.6 - [2 \times (-2 \times 0.5 \times \log 0.5) + 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33)]$
 $= 7.6 - [2 \times 1 + 2.74] = 2.86$

II) Ball Width: $[X_2 = \{6, 22\}]$ $\rightarrow$ Only midpoint is $v'' = 14$ $\rightarrow N^L = 2$, $P^L = \{1, 0, 0\}$
 $N^R = 3$, $P^R = \{0, \frac{2}{3}, \frac{1}{3}\}$

i) $I(X_2 \leq v'' = 14) = 7.6 - [2 \times 0 + 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33)]$
 $= 4.86$

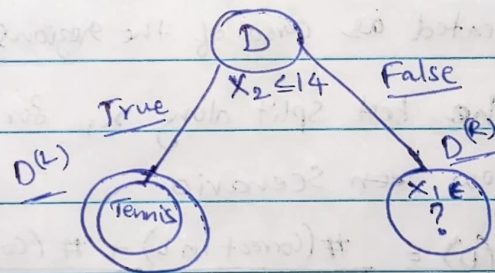
$\therefore$ The initial split that gives highest information gain is
 either Ball Color = {Yellow} & {not Yellow} or (Ball width ≤ 14)

1.2 Construction of entire ~~sub~~ decision tree

⇒ Let's consider the Ball width split as the first one.

⇒ The left subtree doesn't require any further split and can be assigned to Tennis.

⇒ The right subtree would need a split on Ball Color.

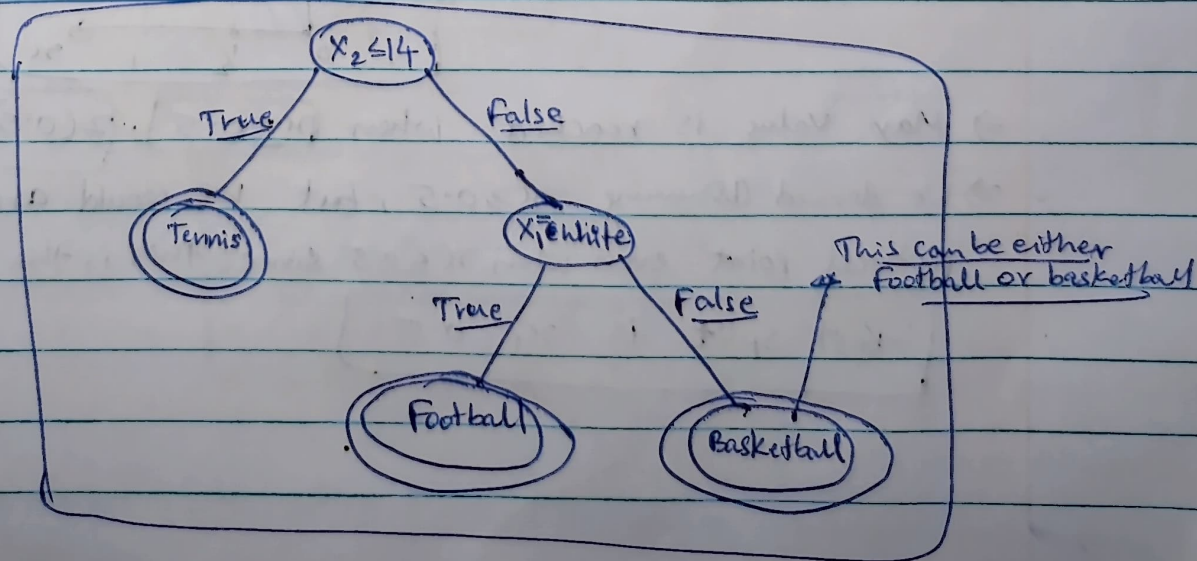


⇒ Let's calculate the information gain with the split $\begin{cases} L \in \text{White} \\ R \in \text{Brown} \end{cases}$.

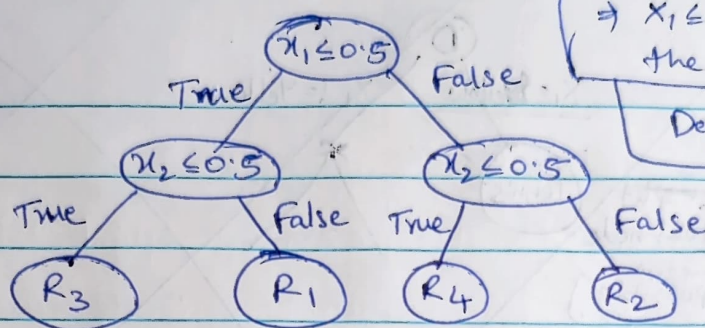
[Ball-width can't be split further, since it's $22 \neq d \in D(R)$.]

$$\begin{aligned} \Rightarrow \text{Info Gain} &= 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33) - [1 \times 0 + 2 \times (-2 \times 0.5 \times \log 0.5)] \\ &= 2.74 - 2 = \boxed{0.74} \end{aligned}$$

The final Decision Tree



2.1) Decision Tree for the given region splits 2



$\Rightarrow x_1 \leq 0.5 \wedge x_2 \leq 0.5$ are the decision splits.

Depth of Tree = 2

2.2) $\Rightarrow$ A metric to match the requirements where 1 is the best metric and 0 is the worst can be ~~something derived from Gini Index~~ accuracy score -

$$\text{Metric} \equiv \frac{\#(\text{Correct Predictions})}{\#(\text{Total Predictions})}$$

[Let's refer to this Metric as $Z(D)$]

It can also be the accuracy score

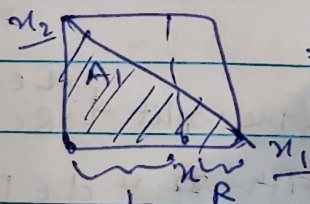
$\Rightarrow$ For the above decision tree it is as below -

$$Z(D) = \frac{1}{4} [Z(R_1) + Z(R_2) + Z(R_3) + Z(R_4)] = \frac{1}{4} [0 + 1 + 0 + \frac{1}{2}] = 0.5$$

[This number is treated as area of the regions, since it's continuous space]

2.3) Let's figure out the best split along x_1 for decision stump (depth=1).

$\Rightarrow$ Consider the below seen scenario -

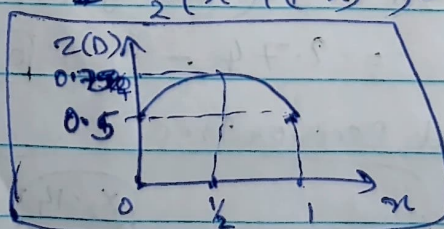


$$\Rightarrow Z(D) = \frac{\#(\text{Correct in L}) + \#(\text{Correct preds in R})}{\#(L+R)}$$

$$= \frac{(x \cdot \frac{1}{2}x^2) + ((1-x) \cdot \frac{1}{2}(1-x)^2)}{1} \quad [\text{Assuming } x \geq 0.5]$$

[Left would predict class-1 and R would predict class-0 (Assuming $x \geq 0.5$)]

$$= 1 - \frac{1}{2}(x^2 + (1-x)^2) = -x^2 + x + 0.5$$



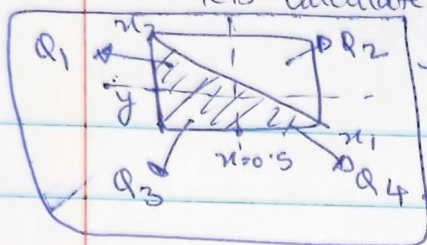
$\Rightarrow$ Max value is reached when $x \leq 0.5$ $\cdot$ $Z(0.5) = 0.75$

$\Rightarrow$ We derived assuming $x \geq 0.5$, but we would arrive at the same optimal point even when $x \leq 0.5$ since this is the limiting case.

$\therefore$ Best split is $x_1 \leq 0.5$

2.4) Next split (on axis x_2): Given $x_1 \leq 0.5$ as the first split,

→ let's calculate the best split of y on axis x_2 .



→ let calculate first for $y \leq 0.5$.

→ Let's call the regions as Q_1, Q_2, Q_3, Q_4 .

(Resultant)
 $\Rightarrow Z(D) = \#(\text{Correct in } Q_1) + \#(\text{Correct in } Q_2) + \#(\text{Correct in } Q_3) + \#(\text{Correct in } Q_4)$
 ~~$\Rightarrow Z(D) = \#(Q_1 + Q_2 + Q_3 + Q_4)$~~

$\Rightarrow Z(D) = \#(\text{Correct in } Q_1) + \#(\text{Correct in } Q_2) + \#(\text{Correct in } Q_3) + \#(\text{Correct in } Q_4)$

[Note that; Q_1 predicts class 1, Q_2 predicts class 0, Q_3 predicts class -1, Q_4 predicts class -1]

$$\Rightarrow Z(D) = \underbrace{\frac{1}{2} \left(\frac{1}{2}((1-y) + (0.5-y)) \right)}_{Q_1} + \underbrace{\left(\frac{1}{2}(1-y) - \frac{1}{2}(0.5-y)^2 \right)}_{Q_2} + \underbrace{(0.5y)}_{Q_3} + \underbrace{\left(\frac{1}{2}(y) - \frac{1}{2}(y)(y) \right)}_{Q_4}$$

$$\Rightarrow Z(D) = \left(\frac{3}{8} - \frac{y}{2} \right) + \left(\frac{1}{2} - \frac{y}{2} - \left(\frac{y^2}{2} - \frac{y}{2} + \frac{1}{8} \right) \right) + \left(\frac{y}{2} \right) + \left(\frac{y}{2} - \frac{y^2}{2} \right)$$

$$Z(D) = -y^2 + y/2 + 0.75$$

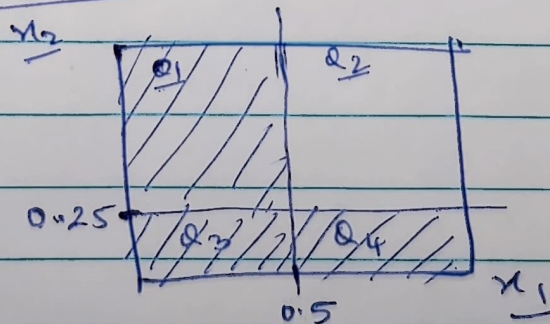
$\Rightarrow \text{Max}^m$ occurs at $y = 0.25$ and $z(0.25) = -\frac{1}{16} + \frac{1}{8} + \frac{6}{8} = \frac{13}{16}$

→ Due to symmetry a split at $y = 0.75$ gives the same z -score.

∴ Next best split would be $x_1 \leq 0.25$ & $x_2 \leq 0.75$

Resultant Decision Tree Regions

Option-1



Option-2

